Zili Tan

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Research Interests

Self-Improving AI Systems, Learning Dynamics, Continual Adaptation, and Multi-Agent Systems.

Academic Background

University of Amsterdam & Vrije Universiteit Amsterdam *09/2025 – 06/2027 (expected)*
MSc in Physics and Astronomy (joint degree), Theoretical Physics track Current average grade: 8.62/10

Sun Yat-sen University *09/2021 – 07/2025*
Bachelor of Science in Physics Average grade: 8.57/10

Publications & Preprints

Meng Bi, Jinhai Yang, **Zili Tan**, Bowei He, Hong Huang, and Xue Liu. *Wright–Gödel Machine: Mitigating Homogenization in Self-Improving Agents.* **Under review at ICLR 2027.**

Research Experience

Continual Learning in Resource-Constrained Agents *10/2026 – Present*
Exploratory research

- Investigating whether prior experience remains useful across changing task streams under memory and compute constraints.
- Built and locally tested a research prototype; prepared a manuscript draft.
- Designed controlled evaluations with matched resource budgets to assess cumulative task performance and the value of prior experience for recurring and held-out tasks.

Agentic AI Workflows for SMEFT Event Generation at the LHC *10/2026 – Present*
MSc Thesis, Advisor: Prof. Wouter Verkerke, Nikhef / University of Amsterdam

- Developed a single-agent workflow in LangGraph that configures MadGraph from physics requests, generates parton-level SMEFT events, and produces kinematic plots.
- Integrated mandatory configuration and sample checks with corrective feedback and review of unresolved choices in the physics setup.
- Evaluated four leading-order tasks in paired tests against independent reference samples: 4/4 produced matching samples with checks and review enabled, versus 0/4 with them disabled.
- Tested Python 3 compatibility across all ten official SMEFTsim UFO variants using 100-event MadGraph runs.
- Published bilingual tutorials on SMEFTsim event generation and detector simulation.

Genetic Homogenization in Self-Improving Coding Agents *08/2026 – 09/2026*
Collaborative project with Meng Bi et al.

- Investigated diversity collapse in recursively self-improving coding agents across Darwin, Huxley, and Mendel Gödel Machine variants.
- Jointly introduced the concept of genetic homogenization to characterize the loss of heritable design diversity during iterative self-improvement.
- Studied the exploration–exploitation tradeoff underlying this collapse and proposed EXP3 as an alternative to Thompson Sampling for maintaining architectural diversity.
- Jointly introduced an evaluation strategy inspired by BAI-MCTS.

Observer-Relative Inflationary Histories in de Sitter JT Gravity *06/2026 – 09/2026*
Advisors: Prof. Ben Freivogel and Prof. Jan Pieter van der Schaar, University of Amsterdam

- Studied de Sitter JT (Jackiw–Teitelboim) gravity coupled to a homogeneous inflaton and a weakly coupled observer.
- Found that the current observer-relative gravitational path integral (GPI) prescription fails to normalize the no-boundary state, suggesting a possible inconsistency in the Hartle–Hawking proposal within the observer-relative Hilbert space.

Lorentzian Gravitational Path Integral in the de Sitter Static Patch *04/2026 – 06/2026*
Advisor: Prof. Jan Pieter van der Schaar, University of Amsterdam

- Reviewed how Lorentzian path-integral approaches can avoid the conformal factor problem.
- Used Claude Code to expand the Einstein–Hilbert action around the D-dimensional de Sitter static patch.
- Obtained the one-loop spectrum of de Sitter space from the Lorentzian gravitational path integral.

Holographic Cosmology and Inflationary Cosmology *10/2022 – 06/2025*
BSc Thesis, Advisor: Prof. Chengcheng Han, Sun Yat-sen University

- Studied the standard cosmological model and inflationary cosmology in Prof. Han's research group.
- Reviewed the holographic principle, holographic entropy bounds, and models proposed by Banks and Fischler.
- Analyzed how holographic models address the problem of low-entropy initial conditions and assessed prospects for testing holographic cosmology through future CMB and primordial gravitational-wave observations.

Competitions & Awards

National Training Program of Innovation for Undergraduates *12/2022 – 03/2024*
Presenter and Team Member

- Reviewed slow-roll models and compared their predicted power spectra with Planck TT observations.
- Final report rated **Excellent** (03/2024).

China Undergraduate Physics Tournament (CUPT) *12/2021 – 10/2022*
Contestant representing SYSU

- Reviewed literature on fluid mechanics and Hamiltonian mechanics; designed experiments and developed theoretical models for physics problems motivated by everyday phenomena; presented results in the Central South Regional and National rounds.
- Received the **National First Prize** at CUPT (10/2022).
- Received the **First Prize** in the Central South Regional Competition (06/2022).

Extracurricular Activities

SYSU Freshman Mentoring Program *09/2024 – 09/2025*
Mentor

- Mentored a group of incoming students throughout their first year, offering guidance on their studies and the transition to university life.

SYSU Torchwood Association of Physics *10/2022 – 10/2024*
Governing Member

- Organized information-sharing seminars and physics contests, including the CUPT Simulation Game.
- Served as a judge in several contests.

Skills

- Python, Mathematica, LaTeX, and Linux.